



NITI Aayog



SECTOR AT A GLANCE 12

Energy Storage in India

Energy in India

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ENERGY STORAGE IN INDIA

ENERGY IN INDIA

The '**Sector-at-a-Glance**' provides an easy-to-read overview and basic understanding of a sector, sub-sector, or thematic area. The facts and insights on the sector are supported by relevant data and best practices from credible government and multilateral sources. The document covers the structure, key trends, opportunities and challenges, government initiatives, and the way forward for the sector.

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Overview

The importance of energy storage technologies has grown significantly in India driven primarily by the increasing focus and ambition of integrating renewable energy rapidly into our energy mix. Energy storage optimises energy systems by storing electricity for later use, serving various applications like grid-level balancing for renewables, electronics, behind-the-meter (BTM) power backup inverters, and supporting the EV sector. In India, commercially viable options include batteries and pumped hydro storage plants, while emerging technologies like flywheels, supercapacitors, and hydrogen offer promising potential. As of 2023, the overall energy storage capacity developed in the country is 4745.60 MW from Hydro Pumped Storage Projects (PSPs) and 39.12 MWh from Battery Energy Storage System (BESS). Energy storage in India witnessed a 23 GWh demand in 2018, 56% of which came from the power backup inverter segment. Furthermore, the EV industry accounted for more than 5 GWh of batteries in 2018.¹

The creation of a conducive energy storage ecosystem is crucial for achieving India's sustainable energy goals and has the potential to strengthen the country's power and mobility sectors.

However, challenges related to low domestic reserves for raw materials like cobalt, and lithium, high import dependency and limited regulatory focus on energy storage impede investments in energy storage systems.

Given these opportunities and challenges, incentivising domestic manufacturing of battery and cells and creating a conducive regulatory environment for energy storage are the major policy focus areas. Several ongoing and recent central government initiatives, including but not limited to the National Programme on Advanced Chemistry Cell (ACC) Battery Storage and National Mission on Transformative Mobility and Storage focus on enhancing India's battery manufacturing capacity alongside ensuring global competitiveness of levelized battery cost. These efforts and several state- and local-level policies and initiatives seek to accelerate India's transition to a clean energy economy while supporting vehicle electrification and renewable energy integration.

Accelerating domestic manufacturing, facilitating deployment of a variety of storage technology solutions, accelerating indigenous R&D, encouraging sustainable practices to limit battery waste, and promoting greater uptake of and investments in renewable energy storage solutions will be critical to continue advancing India's energy storage ambitions. This note presents key trends related to different components of energy storage in India and actions for the way forward.

Figure 1: Key energy storage statistics

47.65 GWh

Projected Pumped Storage Hydropower (PSP) Storage Need in 2026-27

34.72 GWh

Projected Battery Energy Storage System (BESS) Need in 2026-27

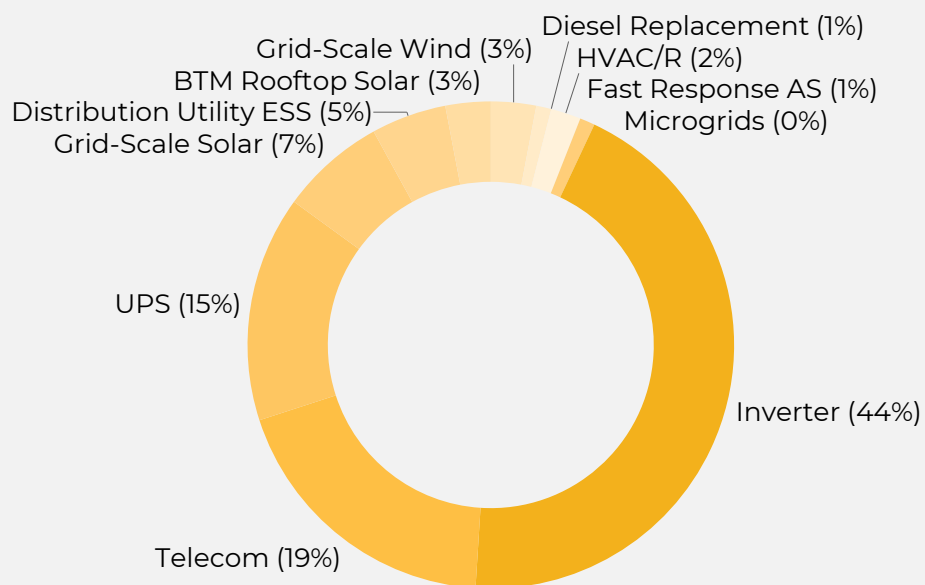
2700 GWh

Estimated energy storage requirements by 2032

Source: India Smart Grid Forum, Energy Storage System Roadmap for India: 2019-2032, 2019; Ministry of New and Renewable Energy, 64.54 billion units electricity produced from Wind Energy during April, 2022-January, 2023 – Union Power & NRE Minister, 2023; Ministry of Power, National Electricity Plan, 2023

¹ India Smart Grid Forum, Energy Storage System Roadmap for India: 2019-2032, 2019; India Energy Storage Alliance, India Stationary Behind-the-meter (BTM) Energy Storage & Railway Battery Market Overview 2021-30, 2022; Ministry of Power, National Electricity Plan (Draft), 2022

Figure 2: 2019-25 energy storage requirement in India



Source: India Smart Grid Forum, Energy Storage System Roadmap for India: 2019-2032, 2019

Opportunities and challenges

Opportunities

This section highlights the outcomes policymakers can achieve by increasing the adoption of energy storage systems. These revolve primarily around reduced curtailment of renewable power, cost-effective electric vehicles, and accelerated transition to clean energy.



Enhancing storage infrastructure to support renewable energy integration:

Integration of huge amounts of variable and intermittent RE threatens grid operations, and energy storage can contribute to greater grid stability with efficient management of this RE influx.² Further, distributed RE systems such as microgrids and lanterns alongside increased off-grid systems, require storage solutions like batteries.³ This Battery Energy Storage Systems (BESS) deployment combined with rooftop solar or microgrids can provide electricity access to remote areas, enhancing their overall development.⁴ A report by IEEFA highlights the enormous opportunity for rooftop solar and BESS to replace diesel gensets, which represent 90 gigawatts (GW) of aggregate capacity in behind-the-meter applications in India.⁵ According to the Central Electricity Authority, as India aims to reach approximately 500 GW of renewable energy capacity by 2030, 34 GW of BESS will be required to stabilise the grid.⁶



Supporting the growth of the Electric Vehicle (EV) market:

In line with the shift to non-fossil fuel-based energy and the launch of the National Mission for Transformative Mobility, a massive boom in EV penetration is expected in India over the next decade. Thus, battery storage systems will become indispensable to support this EV market and smoothly integrate EV charging into grid operations.⁷ In 2018, the EV industry in India consumed over 5 GWh of batteries, which is expected to rise over 36 GWh by 2025.⁸



Boosting domestic manufacturing of batteries:

A domestic battery manufacturing ecosystem with elements like a robust supply chain of cells will be crucial for India to gain an edge in mobility, grid energy storage, and consumer electronics. This will help reduce import costs of li-ion cells, which have increased sevenfold between 2014 and 2020, safeguard against potential supply shocks, and create new jobs. Further, there are opportunities for private players to invest in this sector through arrangements like government partnerships which enable risk-sharing. The government has already started providing incentives for manufacturers through the PLI scheme approved in 2021.⁹

Challenges

This section highlights some of the major challenges and constraints facing energy storage in India. Efforts to resolve such challenges will ensure cost-effective domestic manufacturing of batteries and increase adoption of energy storage systems.

² India Smart Grid Forum, Energy Storage System Roadmap for India: 2019-2032, 2019

³ NITI Aayog and Green Growth Equity Fund Technical Cooperation Facility, Advanced Chemistry Cell Battery Reuse and Recycling Market in India, 2022

⁴ Indian Institute of Management Ahmedabad, International study on financing needs for new age critical clean energy, 2023

⁵ IEEFA, Emerging Technology Trends in the C&I Rooftop solar market in India, 2021

⁶ IEEFA, Flexing India's Energy System Through Market Mechanisms, 2019

⁷ NITI Aayog and Green Growth Equity Fund Technical Cooperation Facility, Advanced Chemistry Cell Battery Reuse and Recycling Market in India, 2022

⁸ India Smart Grid Forum, Energy Storage System Roadmap for India: 2019-2032, 2019

⁹ Cabinet, Cabinet approves Production Linked Incentive scheme "National Programme on Advanced Chemistry Cell Battery Storage", 2021; NITI Aayog and RMI, Need for Advanced Chemistry Cell Energy Storage in India, 2022

Limited domestic reserves, high costs, and currently insufficient technological knowledge affect domestic battery manufacturing and new investments:

India has limited domestic capacity for commercial manufacturing of battery cells. The country has very low domestic reserves of essential components required for manufacturing, like cobalt, and battery-grade graphite (lithium reserves were recently discovered in Jammu & Kashmir).¹⁰ Currently, India imports most of its li-ion cells, and li-ion batteries comprise as high as 30-40% of an EV's cost in India. Further, India has limited technical expertise in this area, and technology patents further inhibit information exchange and technology transfer.¹¹ Finally, the high cost of establishing new storage facilities presents a barrier to new investments in India, as existing financial incentives for storage are limited.¹²



High dependency on battery imports:

In 2022, India's battery import expenditure reached \$1.8 billion, with more than 85% of imports originating from China. India faces various challenges in establishing a stronger grip on the lithium-ion batteries supply chain, including a lack of infrastructure across the entire supply chain and limited experience in deployment.¹³



Limited regulatory focus towards battery use and safety standards:

Some of the current regulations do not allow the provision of storage services and restrict earning revenue. Such wholesale and retail barriers make it difficult for storage solutions like batteries to compete in the market and hamper maximising investment return.¹⁴ Further, concerning the safety of use, India has adopted specific international standards for energy storage systems. However, these are not contextualised to the country's operating and tropical climatic conditions, which can lead to accidents like battery fires and poor customer trust.¹⁵



Limited waste management strategies:

India's surge in the adoption of energy storage solutions, specifically li-ion batteries (LIB), raises concerns about the potential overwhelming waste generation. While the Ministry of Environment, Forest, and Climate Change introduced the Battery Waste Management Rules in 2022, there is a need to implement and develop a robust recycling system, as past regulations did not address LIB recycling adequately. Without effective implementation it can have harmful environmental impacts, such as soil and groundwater contamination.¹⁶



¹⁰ NITI Aayog and RMI, Need for Advanced Chemistry Cell Energy Storage in India, 2022; Ministry of Mines, Geological Survey of India Finds Lithium and Gold Deposits, 2023

¹¹ NITI Aayog and RMI, Need for Advanced Chemistry Cell Energy Storage in India, 2022; Ministry of Heavy Industries, Evaluation of Electric Vehicle (EV) Policy, 2023

¹² NREL, Policy and Regulatory Readiness for Utility-Scale Energy Storage: India, accessed June 2023

¹³ NITI Aayog and RMI India, Need for Advanced Chemistry Cell Energy Storage in India, February 2022; CEEW, How can India Indigenise Lithium-Ion Battery Manufacturing?, 2023

¹⁴ IEA, Renewables Integration in India, 2021

¹⁵ IESA and WBCSD, Policy and Regulatory Framework to Accelerate Energy Storage Deployment in India, accessed June 2023

¹⁶ NITI Aayog and Green Growth Equity Fund Technical Cooperation Facility, Advanced Chemistry Cell Battery Reuse and Recycling Market in India, 2022; Ministry of Environment, Forest and Climate Change, Government notifies Battery Waste Management Rules, 2022

Institutional framework and policy landscape –

Key ministries and departments

This section highlights the major central government bodies and national-level institutions that oversee energy storage in the country.



NITI Aayog

The NITI Aayog serves as the apex public policy think tank of the Government of India. Within the energy vertical, deals with five key sectors: power, coal, petroleum and natural gas, new and renewable energy, and atomic power. It conducts evaluations of various storage technologies, analyzes policies, and takes steps to enhance energy planning, data management, and other measures to incentivize the development and deployment of energy storage systems.



Ministry of Power

The ministry administers, formulates, and implements power sector-related policies and provides R&D and technical assistance relating to grid integration, transmission system networks and distribution systems in the States/UTs.



Ministry of New and Renewable Energy (MNRE)

The ministry develops renewable and clean energy technologies, systems, products, and services in line with global standards and deploys these for energy applications for sectors like power, transport and support country's objective of achieving energy security.



Bureau of Energy Efficiency

The bureau is the nodal authority for all GH2 projects in India, assists in developing policies and strategies to promote energy efficiency and conservation and utilises existing resources and infrastructure in performing functions under the Energy Conservation Act.



Indian Renewable Energy Development Agency

The agency finances and supports self-sustainable investments in energy generation from renewable sources, and carry out promotion of energy efficiency and sustainable environmental technologies.



Ministry of Heavy Industries

The Ministry of Heavy Industries oversees the development and growth of heavy industries in India. Its policies and initiatives impact the manufacturing and deployment of storage technologies.



Solar Energy Corporation of India (SECI)

SECI, a government of India enterprise under MNRE, facilitates the development of India's solar energy market. They implement large-scale solar power projects, promoting grid-connected and off-grid solar power solutions



Power Finance Corporation (PFC)

A public sector financial institution under the Ministry of Power, PFC finances the power sector through loans and investments. They play a crucial role in funding generation, transmission, and distribution projects, promoting sector growth.



REC Limited

A public sector financial institution under the Ministry of Power, REC primarily focuses on rural electrification and enhancing power infrastructure in remote areas. It finances and supports electrification projects, grid expansion, and distribution network improvement.



Central Electricity Authority (CEA)

Functioning under the Ministry of Power, CEA acts as the apex technical and planning body for the power sector in India. As the technical planning and regulatory body, CEA is responsible for developing grid integration standards and regulations for energy storage systems. Additionally, CEA plays a role in forecasting future energy storage requirements and shaping national policies in this domain.

Major policies and programs

This section highlights existing policies related to energy storage and their most prominent features. A more comprehensive list of policies is provided in the policy repository of the NITI Knowledge Platform.



National Energy Storage Mission, 2018¹⁷

In 2018, an expert committee under MNRE proposed a draft National Energy Storage Mission. The mission aims to achieve leadership in the energy storage sector by promoting manufacturing, deployment, cost reduction, and innovation.



National Programme on Advanced Chemistry Cell (ACC) Battery Storage, 2022¹⁸

This programme is a Production Linked Incentive (PLI) scheme implemented by the Ministry of Heavy Industry, which aims to enhance India's manufacturing capabilities, alongside ensuring global competitiveness of the country's levelised battery cost. The objective is to achieve a manufacturing capacity of 50 GWh of ACC. This scheme is technology agnostic, thus enabling beneficiary firms to choose suitable advanced technology and corresponding requirements.



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National Mission on Transformative Mobility and Battery Storage, 2019¹⁹

The mission aims to promote environmentally friendly and sustainable mobility initiatives. This includes implementing a Phased Manufacturing Programme (PMP) to help establish battery and cell manufacturing giga plants that are large-scale and export competitive. Furthermore, another PMP encourages localised production throughout the entire value chain of Electric Vehicles (EVs).



Initiatives to promote growth of energy storage technologies, 2022-23²⁰

In the Union Budget 2023-24, the Ministry of Power introduced a Scheme for Viability Gap Funding to develop 4,000 MWh Battery Energy Storage Systems (BESS). The Ministry of Power has also introduced a waiver of Inter State Transmission System (ISTS) Charges on Hydro Pumped Storage Projects (PSP) for 25 years and BESS Projects for 12 years. Other initiatives include:

- Waiver of ISTS charges on Hydro Pumped Storage Projects
- Legal status and bidding guidelines for Energy Storage Systems
- Flexibility in generation and scheduling of Thermal/Hydro power stations through bundling with Renewable Energy and Storage Power



¹⁷ Ministry of New and Renewable Energy, Government of India has been making concerted efforts for Developing Energy Storage Technologies, 2019

¹⁸ Ministry of Heavy Industries, Three Companies signed Program Agreement under (PLI) Scheme for Advanced Chemistry Cell (ACC) Battery Storage, 2022

¹⁹ Cabinet, Mobility Solutions get a boost, 2019

²⁰ Ministry of Power, Energy storage technology, 2022; Ministry of New and Renewable Energy, Renewable energy and storage issues, 2023; Ministry of Power, National Framework for Promoting Energy storage systems, 2023

- Inclusion of “Energy storage systems” in the Harmonized Master List of Infrastructure subsectors
- National Framework for promoting Energy storage systems with an objective of round the clock renewable power.

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Way forward

This section highlights considerations for policymakers seeking to advance policy formulation and implementation in this sector. These are based on reports and materials prepared by key government bodies and expert institutions like the NITI Aayog and the Ministry of Power.

 Enhance domestic manufacturing capacity	Conduct sectoral assessments - Collaborate with industry stakeholders, research institutions, and state governments to identify specific sectors with high energy storage demand. Prioritize sectors such as renewable energy integration, electric mobility, and microgrids. Promote advanced chemistry cell (ACC) battery manufacturing: - Establish a supportive ecosystem for gigafactories that manufacture ACC batteries and energy storage systems (ESS). Incentivize investments in R&D, skill development, and infrastructure. Leverage and advance existing initiatives such as production-linked incentives and the Make in India campaign to attract global battery manufacturers and joint ventures with Indian firms.
 Facilitate technology agnostic deployment	Advance resource adequacy planning - Collaborate with grid operators and utilities to develop resource adequacy plans that integrate energy storage. Assess grid requirements, peak demand, and load profiles to determine optimal storage capacity. Provide regulatory support for enhancing grid connectivity and access - Streamline regulatory processes for grid connectivity of ESS projects. Provide clear guidelines for grid integration, including technical standards and interconnection procedures. Technology-neutral bidding guidelines - Develop bidding guidelines that allow flexibility in technology choice (e.g., lithium-ion, flow batteries, compressed air energy storage). Encourage competition and cost-effectiveness. Recent tenders and auctions have adopted this approach.
 Promote Research and Development	Enable indigenous technology innovation with academia-industry collaborations - Establish research centers and consortia focused on energy storage technologies. Encourage collaboration between academia, industry, and startups to develop indigenous solutions. Finance pilot projects - Fund pilot projects to validate innovative storage technologies in real-world scenarios. Monitor performance, reliability, and cost-effectiveness.



Encourage sustainable practices and recycling

Advance a circular economy approach

- Develop policies that promote recycling and reuse of batteries. Encourage manufacturers to design products with end-of-life considerations. Implement extended producer responsibility (EPR) programs. Existing EPR programs in India can be leveraged for this purpose.

Develop and deploy monitoring and evaluation frameworks for energy storage

- Establish a monitoring framework to track the environmental impact of energy storage deployment. Evaluate the lifecycle emissions, resource consumption, and waste management practices. Environmental impact assessments (EIAs) play a crucial role in this process.



Support renewable energy storage

Support solar and wind paired storage

- Encourage the pairing of battery storage with solar or wind generation to ensure a steady supply of power in spite of intermittent generation. This can be achieved through policy incentives and regulatory support.

Enable greater grid integration of renewable energy storage

- Develop clear guidelines and standards for the integration of renewable energy storage into the grid. This will help to ensure the stability and reliability of the power supply.

Incentivize investment in renewable energy storage

- Provide financial incentives and policy support to encourage investment in renewable energy storage. This could include tax credits, grants, or low-interest loans for projects that incorporate battery storage with solar or wind generation.

Recommended readings

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